JEE Mathematics: Quick Review with Grading and Practice Map - Inverse Trigonometric Function (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (58 questions; a question that appears in two of these is counted once): High = 5 or more questions, Medium = 1 to 4 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction of ITF

Question 01

Define inverse of a trigonometric function.

Model Answer

A trigonometric function is many-one on R; restricted to a principal interval where it is one-one and onto, it has an inverse. $y = \sin x$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $y \in [-1, 1]$ gives $\sin^{-1} : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
Ex: $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$ (not $\frac{5\pi}{6}$, which is outside the principal range).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

What do you mean by arc(sin(x))?

Model Answer

$\arcsin x = \sin^{-1} x =$ the angle in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ whose sine is x. Note $\sin^{-1} x \neq \dfrac{1}{\sin x}$.

Ex: $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Domain and Range of 6 ITFs

Question 03

What is domain and range of all 6 ITFs?

Model Answer

$\sin^{-1} x$: $[-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$; $\cos^{-1} x$: $[-1, 1] \to [0, \pi]$; $\tan^{-1} x$: $R \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.
$\cot^{-1} x$: $R \to (0, \pi)$; $\sec^{-1} x$: $|x| \geq 1 \to [0, \pi] - \left\{\frac{\pi}{2}\right\}$; $\operatorname{cosec}^{-1} x$: $|x| \geq 1 \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$.
For $\sin^{-1} u$, $\cos^{-1} u$ put $-1 \leq u \leq 1$; for $\sec^{-1} u$, $\operatorname{cosec}^{-1} u$ put $|u| \geq 1$. Ex: $\sin^{-1}(\ln x)$: $-1 \leq \ln x \leq 1 \Rightarrow x \in \left[\frac{1}{e}, e\right]$.

Grade - High

Practice map, main concept: Sheet: J-II 1-2, 4-6, 33-34, 41; J-M 3-4, 8, 11, 17, 21, 23; J-A 7; C-I 10, 12 | DPP: 6(1, 3-4, 7-16) | Question Bank: 72

Also used in: Sheet: J-I 9, 20; J-M 1; C-II 5, 8, 17 | DPP: 6(17); 8(2)

Question 04
What are functions of first family?

Model Answer
$\sin^{-1} x$, $\tan^{-1} x$, $\text{cosec}^{-1} x$: range symmetric about 0 (in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$); all are odd: $f(-x) = -f(x)$.
Ex: $\tan^{-1}(-2) = -\tan^{-1} 2$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 05
What are the functions of second family?

Model Answer
$\cos^{-1} x$, $\cot^{-1} x$, $\sec^{-1} x$: range in $[0, \pi]$; neither even nor odd, $f(x) + f(-x) = \pi$.
Ex: $\cos^{-1}\left(-\frac{1}{4}\right) = \pi - \cos^{-1}\frac{1}{4}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Graph of 6 ITFs

Question 06
Plot graph of all six ITFs.

Model Answer
Each graph is the mirror image of the restricted trigonometric graph in $y = x$. $\sin^{-1} x$: from $\left(-1, -\frac{\pi}{2}\right)$ to $\left(1, \frac{\pi}{2}\right)$ through O; $\cos^{-1} x$: from $(-1, \pi)$ to $(1, 0)$ through $\left(0, \frac{\pi}{2}\right)$.
$\tan^{-1} x$: asymptotes $y = \pm\frac{\pi}{2}$; $\cot^{-1} x$: asymptotes $y = 0$, $y = \pi$, passes $\left(0, \frac{\pi}{2}\right)$; $\sec^{-1} x$, $\text{cosec}^{-1} x$: two branches for $|x| \geq 1$ with asymptote $y = \frac{\pi}{2}$, resp. $y = 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 07
Find derivative of all six ITFs.

Model Answer

$$\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\tan^{-1} x = \frac{1}{1+x^2}, \quad \frac{d}{dx}\sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$$

The co-functions have the same derivatives with a minus sign: $\cos^{-1} x \to -\frac{1}{\sqrt{1-x^2}}$, $\cot^{-1} x \to -\frac{1}{1+x^2}$, $\text{cosec}^{-1} x \to -\frac{1}{|x|\sqrt{x^2-1}}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 08
Comment on monotonic nature of first four ITFs.

Model Answer
$\sin^{-1} x$, $\tan^{-1} x$ are increasing; $\cos^{-1} x$, $\cot^{-1} x$ are decreasing on their domains.
Ex: $\sin^{-1} x + \tan^{-1} x$ is increasing on $[-1, 1]$, so its range is $\left[-\frac{3\pi}{4}, \frac{3\pi}{4}\right]$.

Grade - Medium
Practice map, main concept: Sheet: J-I 3; J-M 7; J-A 4; C-II 18
Also used in: DPP: 7(13)

Question 09
Comment on odd/even nature of first four ITFs.

Model Answer
$\sin^{-1} x$, $\tan^{-1} x$ are odd; $\cos^{-1} x$, $\cot^{-1} x$ are neither even nor odd ($f(x) + f(-x) = \pi$). All four are bounded and aperiodic.

Grade - Low
Practice map, main concept: Sheet: C-II 17
Also used in: Sheet: J-II 33

Question 10
Comment on concavity of graphs of first four ITFs.

Model Answer
$\sin^{-1} x$, $\cot^{-1} x$: concave up for $x > 0$, concave down for $x < 0$. $\cos^{-1} x$, $\tan^{-1} x$: concave up for $x < 0$, concave down for $x > 0$. $x = 0$ is the point of inflection for all four.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 11
Comment on maximum/minimum value of y for first four ITFs.

Model Answer
$\sin^{-1} x$: max $\frac{\pi}{2}$ at $x = 1$, min $-\frac{\pi}{2}$ at $x = -1$; $\cos^{-1} x$: max π at $x = -1$, min 0 at $x = 1$. $\tan^{-1} x$, $\cot^{-1} x$ have no max/min (the bounds are not attained).
Ex: $\sum_{i=1}^{20} \sin^{-1} x_i = 10\pi \Rightarrow$ every $x_i = 1$, so $\sum x_i = 20$.

Grade - Low
Practice map, main concept: Sheet: J-I 1-2 | DPP: 6(2, 5)
Also used in: Sheet: J-I 3, 8 | DPP: 6(12); 8(3)

Question 12
How to convert negative argument of an ITF expression to positive?

Model Answer

$$\sin^{-1}(-x) = -\sin^{-1} x, \quad \tan^{-1}(-x) = -\tan^{-1} x, \quad \operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$$
$$\cos^{-1}(-x) = \pi - \cos^{-1} x, \quad \cot^{-1}(-x) = \pi - \cot^{-1} x, \quad \sec^{-1}(-x) = \pi - \sec^{-1} x$$

Ex: $\cot^{-1}(-\sqrt{3}) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 11; J-II 30, 37; J-M 9 | DPP: 7(11)

Question 13
How to solve ITF inequality if sin or tan is involved?

Model Answer
Increasing functions keep the sign: $\sin^{-1} x > \sin^{-1} y \iff 1 \geq x > y \geq -1$; $\tan^{-1} x > \tan^{-1} y \iff x > y$. Always add the domain conditions.
Ex: $\sin^{-1} x > \sin^{-1}(2x-1)$: $x > 2x-1$, $x \in [-1,1]$, $2x-1 \in [-1,1] \Rightarrow x \in [0,1)$.

Grade - **Low**
Practice map, main concept: Sheet: C-I 5
Also used in: Sheet: C-II 12-13

Question 14
How to solve ITF inequality if cos or cot is involved?

Model Answer
Decreasing functions reverse the sign: $\cos^{-1} x > \cos^{-1} y \iff -1 \leq x < y \leq 1$; $\cot^{-1} x > \cot^{-1} y \iff x < y$.
Ex: $\cos^{-1} x > \cos^{-1} x^2$: $x < x^2$, $x \in [-1,1] \Rightarrow x \in [-1,0)$.

Grade - **Low**
Practice map, main concept: Sheet: C-II 12 | DPP: 7(7)
Also used in: Sheet: C-II 13 | DPP: 8(4)

Question 15
How to solve inequality of type ITF expression $>/<$ constant?

Model Answer
If the constant lies in the range, take the trigonometric function of both sides (keep the sign for $\sin^{-1}, \tan^{-1}$, reverse it for $\cos^{-1}, \cot^{-1}$) and intersect with the domain. A quadratic in $\sin^{-1} x$ etc.: solve for $t = \sin^{-1} x$ first, with t in the range.
Ex: $\sin^{-1} x > \frac{\pi}{4} \Rightarrow x \in \left(\frac{1}{\sqrt{2}}, 1\right]$; $\cot^{-1} x < \frac{\pi}{6} \Rightarrow x > \sqrt{3}$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 16
How to solve inequality of type ITF expression $>/<$ constant if constant is not in principal domain of ITF?

Model Answer
Compare the constant with the whole range: the inequality is then true for the entire domain or for no x.
Ex: $\sin^{-1} x > \frac{2\pi}{3}$: no x; $\sin^{-1} x < \frac{2\pi}{3}$: $x \in [-1,1]$; $\cot^{-1} x > -\frac{\pi}{3}$: $x \in R$.

Grade - **Low**
Practice map, main concept: Sheet: C-II 13
Also used in: none

Evaluation of inverse trigonometric expression

Question 17
How to convert one inverse to another inverse?

Model Answer
Put $\theta = \tan^{-1} x$ (etc.), draw the right triangle with the matching sides and read off any ratio; use the sign of θ's quadrant. For $f(2\theta)$ use double-angle formulas.

$$\tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1+x^2}} = \cos^{-1} \frac{1}{\sqrt{1+x^2}} \quad (x \geq 0)$$

Ex: $\cos(\tan^{-1} 2) = \frac{1}{\sqrt{5}}$; $\sin(2\tan^{-1} 2) = \frac{2 \cdot 2}{1+4} = \frac{4}{5}$; $\cos(2\tan^{-1} 3) = \frac{1-9}{1+9} = -\frac{4}{5}$.

Grade - High
Practice map, main concept: Sheet: J-I 4, 7, 11; J-II 3, 7, 11-12, 30, 43; J-M 12-14, 18, 24-25; J-A 2, 8-10, 13; C-I 2; C-II 3, 8 | DPP: 6(6); 7(1, 3, 5, 8, 11); 8(1) | Question Bank: 78, 81, 86
Also used in: Sheet: J-M 2, 20; J-A 4; C-I 6; C-II 10 | DPP: 7(2) | Question Bank: 75, 82

Property 1 of ITF

Question 18
Write property 1.

Model Answer
$\sin(\sin^{-1} x) = x$, $\cos(\cos^{-1} x) = x$ for $x \in [-1, 1]$; $\tan(\tan^{-1} x) = x$, $\cot(\cot^{-1} x) = x$ for $x \in R$; $\sec(\sec^{-1} x) = x$, $\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x$ for $|x| \geq 1$. Graph: part of $y = x$ over the domain only.
Ex: $\sin(\sin^{-1} 0.3) = 0.3$, but $\sin(\sin^{-1} 2)$ is not defined.

Grade - Low
Practice map, main concept: Sheet: J-II 29; C-II 14-16
Also used in: Sheet: J-II 36

Property 2 of ITF

Question 19
How to define $\sin^{-1}(\sin x)$, $\cos^{-1}(\cos x)$, $\tan^{-1}(\tan x)$, $\cot^{-1}(\cot x)$ conceptually?

Model Answer
$y = \sin^{-1}(\sin x)$ is the angle in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ with $\sin y = \sin x$; from $x = n\pi + (-1)^n y$ pick the y in the principal range. It equals x only when x itself is in the principal range. Same idea for cos $(x = 2n\pi \pm y)$, tan, cot $(x = n\pi + y)$.
Ex: $\sin^{-1}\left(\sin \frac{2\pi}{3}\right) = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 7; C-II 19

Question 20
How to apply trick to define $\sin^{-1}(\sin x)$?

Model Answer
Round x to the nearest multiple $n\pi$. n even: $\sin^{-1}(\sin x) = x - n\pi$; n odd: $\sin^{-1}(\sin x) = n\pi - x$.
Ex: $\sin^{-1}(\sin 6) = 6 - 2\pi$; $\sin^{-1}(\sin 9) = 3\pi - 9$.

Grade - Medium
Practice map, main concept: Sheet: C-I 1; C-II 19 | Question Bank: 84
Also used in: Question Bank: 80

Question 21
Plot graph of $y = \sin^{-1}(\sin x)$.

Model Answer
Zig-zag between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, slopes ± 1: $y = x$ on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $y = \pi - x$ on $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$, ...; period 2π, odd, peaks at $x = \frac{\pi}{2} + 2k\pi$.

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: J-II 42

Question 22
How to apply trick to define $\cos^{-1}(\cos x)$?

Model Answer
Round x to the nearest even multiple $2k\pi$: $\cos^{-1}(\cos x) = |x - 2k\pi|$ (i.e. $x - 2k\pi$ if $x > 2k\pi$, $2k\pi - x$ if $x < 2k\pi$).
Ex: $\cos^{-1}(\cos 5) = 2\pi - 5$; $\cos^{-1}(\cos(-5)) = 2\pi - 5$.

Grade - **Medium**
Practice map, main concept: none
Also used in: Sheet: C-I 13 | Question Bank: 80

Question 23
Plot graph of $y = \cos^{-1}(\cos x)$.

Model Answer
Triangle wave between 0 and π: $y = |x|$ on $[-\pi, \pi]$, $y = 2\pi - x$ on $[\pi, 2\pi]$, ...; period 2π, even. Count solutions of $\cos^{-1}(\cos x) = g(x)$ by intersecting with this graph.

Grade - **Medium**
Practice map, main concept: Sheet: J-A 3; C-II 2
Also used in: Sheet: J-II 42

Question 24
How to apply trick to define $\tan^{-1}(\tan x)$?

Model Answer
Round x to the nearest multiple $n\pi$: $\tan^{-1}(\tan x) = x - n\pi$ ($x \neq$ odd multiple of $\frac{\pi}{2}$).
Ex: $\tan^{-1}(\tan 12) = 12 - 4\pi$.

Grade - **Medium**
Practice map, main concept: Sheet: J-II 37 | DPP: 7(9)
Also used in: Sheet: J-M 13; J-A 11, 14

Question 25
Plot graph of $y = \tan^{-1}(\tan x)$.

Model Answer
Saw-tooth: parallel segments of slope 1, $y = x - n\pi$ on $\left(n\pi - \frac{\pi}{2}, n\pi + \frac{\pi}{2}\right)$; period π, odd, breaks at odd multiples of $\frac{\pi}{2}$.

Grade - **Medium**
Practice map, main concept: Sheet: J-A 11
Also used in: none

Question 26
How to apply trick to define $\cot^{-1}(\cot x)$?

Model Answer
Take the integral part of $\frac{x}{\pi}$: $\cot^{-1}(\cot x) = x - \left[\frac{x}{\pi}\right]\pi$ $(x \neq n\pi)$.

Ex: $\cot^{-1}(\cot(-6)) = -6 + 2\pi$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 27
Plot graph of $y = \cot^{-1}(\cot x)$.

Model Answer
Saw-tooth of slope 1 between 0 and π: $y = x - n\pi$ on $(n\pi, (n+1)\pi)$; period π, neither even nor odd, not defined at $x = n\pi$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 28
Plot graph of $y = \sec^{-1}(\sec x)$.

Model Answer
Same as $y = \cos^{-1}(\cos x)$ (triangle wave 0 to π, period 2π, even) with the points $x = (2n+1)\frac{\pi}{2}$ removed.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 29
Plot graph of $y = \text{cosec}^{-1}(\text{cosec}x)$.

Model Answer
Same as $y = \sin^{-1}(\sin x)$ (zig-zag $-\frac{\pi}{2}$ to $\frac{\pi}{2}$, period 2π, odd) with the points $x = n\pi$ removed.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 30
How to define $\sin^{-1}(\sin x)$, $\cos^{-1}(\cos x)$, $\tan^{-1}(\tan x)$, $\cot^{-1}(\cot x)$ for any interval of x?

Model Answer
Split the interval at the turning points ($\frac{\pi}{2}$-multiples for sin, π-multiples for cos, cot; odd $\frac{\pi}{2}$ for tan) and use the trick on each piece. For $f(kx)$ first find the interval of kx.

Ex: $x \in \left[\frac{\pi}{2}, \pi\right] \Rightarrow 2x \in [\pi, 2\pi]$, so $\cos^{-1}(\cos 2x) = 2\pi - 2x$.

Grade - Medium
Practice map, main concept: Sheet: J-II 8, 36, 42; J-M 9, 15; C-I 9, 13 | Question Bank: 80, 82
Also used in: none

Simplification by trigonometric substitutions

Question 31

How to solve questions of simplification by trigonometric substitutions?

Model Answer

Put $x = \sin\theta$ ($\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$), $x = \cos\theta$ ($\theta \in [0, \pi]$) or $x = \tan\theta$ ($\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$); simplify to $f^{-1}(f(k\theta))$ and apply property 2 according to the interval of $k\theta$.

$$\sin^{-1}\frac{2x}{1+x^2} = \begin{cases} 2\tan^{-1}x, & |x| \le 1 \\ \pi - 2\tan^{-1}x, & x > 1 \\ -\pi - 2\tan^{-1}x, & x < -1 \end{cases} \qquad \cos^{-1}\frac{1-x^2}{1+x^2} = \begin{cases} 2\tan^{-1}x, & x \ge 0 \\ -2\tan^{-1}x, & x < 0 \end{cases}$$

$\sin^{-1}(3x-4x^3) = 3\sin^{-1}x \ \left(|x| \le \frac{1}{2}\right)$, $\cos^{-1}(2x^2-1) = 2\cos^{-1}x\ (0 \le x \le 1)$, $\tan^{-1}\frac{2x}{1-x^2} = 2\tan^{-1}x\ (|x| <$

Ex: for $x > 1$, $\sin^{-1}\frac{2x}{1+x^2} + 2\tan^{-1}x = \pi$.

Grade - High

Practice map, main concept: Sheet: J-II 9, 24-28; C-I 7; C-II 9 | DPP: 7(12-13) | Question Bank: 71, 75

Also used in: Sheet: J-II 21, 38-39; J-M 18, 24; J-A 10, 12, 14; C-I 4-5 | DPP: 7(4-5) | Question Bank: 85

Property 4 of ITF

Question 32

How to convert $\sin^{-1}$ to $\cos^{-1}$ and vice versa?

Model Answer

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}, \quad -1 \le x \le 1$$

Ex: $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3} \Rightarrow \cos^{-1}x + \cos^{-1}y = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$; $4\sin^{-1}x + \cos^{-1}x = \pi \Rightarrow 3\sin^{-1}x = \frac{\pi}{2} \Rightarrow x = \frac{1}{2}$.

Grade - High

Practice map, main concept: Sheet: J-I 8-10; J-II 31, 44; J-M 6, 20 | DPP: 7(2); 8(3-5) | Question Bank: 88

Also used in: Sheet: J-I 14; J-II 12, 19-20; J-M 22; J-A 5 | DPP: 6(4); 8(7) | Question Bank: 77, 84, 87, 89

Question 33

How to convert $\tan^{-1}$ to $\cot^{-1}$ and vice versa?

Model Answer

$$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}, \quad x \in R$$

Ex: $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$ gives $x = -1$ $(\frac{\pi^2}{16} + \frac{9\pi^2}{16})$.

Grade - Low

Practice map, main concept: Sheet: J-I 18; C-I 11

Also used in: none

Question 34
How to convert $\sec^{-1}$ to $\operatorname{cosec}^{-1}$ and vice versa?

Model Answer

$$\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}, \quad |x| \geq 1$$

Ex: $\cot^{-1} x + \sec^{-1} x + \operatorname{cosec}^{-1} x = \cot^{-1} x + \frac{\pi}{2}$ for $|x| \geq 1$.

Grade - Medium
Practice map, main concept: Sheet: J-II 32 | DPP: 8(2)
Also used in: Question Bank: 74

Question 35
How to find the range or extreme values of expressions like $(\sin^{-1} x)^2 + (\cos^{-1} x)^2$ or $(\tan^{-1} x)^3 + (\cot^{-1} x)^3$?

Model Answer
Put $t = \sin^{-1} x$ (or $\tan^{-1} x$), the other one is $\frac{\pi}{2} - t$; the expression becomes a polynomial in t; find its range for t in the principal range (vertex of the quadratic, end points).

$$(\sin^{-1} x)^2 + (\cos^{-1} x)^2 = 2\left(t - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8} \in \left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]$$

Ex: $(\tan^{-1} x)^3 + (\cot^{-1} x)^3$, $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$: range $\left[\frac{\pi^3}{32}, \frac{7\pi^3}{8}\right)$.

Grade - Medium
Practice map, main concept: Sheet: J-M 10 | Question Bank: 74
Also used in: none

Property 5 of ITF

Question 36
How to convert $\operatorname{cosec}^{-1}$ to $\sin^{-1}$?

Model Answer

$$\operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}, \quad |x| \geq 1$$

Ex: $\operatorname{cosec}^{-1} 2 = \sin^{-1} \frac{1}{2} = \frac{\pi}{6}$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 8(14)

Question 37
How to convert $\sec^{-1}$ to $\cos^{-1}$?

Model Answer

$$\sec^{-1} x = \cos^{-1} \frac{1}{x}, \quad |x| \geq 1$$

Ex: $\sec^{-1}(-2) = \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 19

Question 38
How to convert $\cot^{-1}$ to $\tan^{-1}$?

Model Answer

$$\cot^{-1} x = \begin{cases} \tan^{-1} \dfrac{1}{x}, & x > 0 \\ \pi + \tan^{-1} \dfrac{1}{x}, & x < 0 \end{cases}$$

Ex: $\cot^{-1}(-1) = \pi + \tan^{-1}(-1) = \frac{3\pi}{4}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-II 15-17, 35; J-A 1, 12 | Question Bank: 76

Addition or Subtraction of angles

Question 39
How to add two angles?

Model Answer

$$\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \dfrac{x+y}{1-xy}, & xy < 1 \\ \pi + \tan^{-1} \dfrac{x+y}{1-xy}, & x > 0,\ y > 0,\ xy > 1 \\ \dfrac{\pi}{2}, & x > 0,\ y > 0,\ xy = 1 \end{cases}$$

$\sin^{-1} x + \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$ for $x, y \geq 0$, $x^2 + y^2 \leq 1$. Three angles: $\tan(A + B + C) = \dfrac{S_1 - S_3}{1 - S_2}$ with $S_1 = x + y + z$, $S_2 = xy + yz + zx$, $S_3 = xyz$.

Ex: $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8} = \tan^{-1}\frac{2}{3} + \tan^{-1}\frac{1}{5} = \tan^{-1} 1 = \frac{\pi}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 5-6, 13, 15; J-II 13-15, 17-18, 35, 38-40; J-M 2, 19; C-I 8; C-II 4 | DPP: 7(4) | Question Bank: 79

Also used in: Sheet: J-II 21; J-A 2; C-II 1 | DPP: 7(1); 8(8-9)

Question 40
How to subtract two angles?

Model Answer

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}, \quad xy > -1$$

$\cos^{-1} x - \cos^{-1} y = \cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right)$ for $0 \leq x \leq y \leq 1$. In general take cos (or tan) of both sides and use $\cos(A - B)$.

Ex: $\tan^{-1} 3 - \tan^{-1}\frac{1}{2} = \tan^{-1}\frac{3-\frac{1}{2}}{1+\frac{3}{2}} = \tan^{-1} 1 = \frac{\pi}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 12, 14, 17, 21; J-II 10, 16 | Question Bank: 76-77

Also used in: Sheet: J-I 19; J-A 13 | DPP: 7(8); 8(6)

Question 41

How to deal factor of 1/2 in addition/subtraction of angles?

Model Answer

Put $\theta = \cos^{-1} x$ (or $\tan^{-1} x$), so the term is $\frac{\theta}{2}$, and use half-angle formulas; or write $\frac{1}{2}\cos^{-1} x$ as one $\tan^{-1}$ first.

$$\tan\left(\tfrac{1}{2}\cos^{-1} x\right) = \sqrt{\frac{1-x}{1+x}}, \qquad \cos\left(\tfrac{1}{2}\cos^{-1} x\right) = \sqrt{\frac{1+x}{2}}$$

Ex: $\cos\left(\frac{1}{2}\cos^{-1}\frac{1}{8}\right) = \sqrt{\frac{9/8}{2}} = \frac{3}{4}$.

Grade - **Medium**

Practice map, main concept: Sheet: C-II 10 | DPP: 7(6, 10)

Also used in: Sheet: J-II 14; J-M 18, 24

ITF Equations

Question 42

How to solve ITF equations?

Model Answer

(i) Use $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ etc. to get one function; (ii) combine angles with the addition formulas, or take sin/cos/tan of both sides; (iii) solve the algebraic equation; (iv) check each root in the original equation. Counting roots: compare ranges or draw graphs.

Ex: $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4} \Rightarrow \frac{5x}{1-6x^2} = 1 \Rightarrow 6x^2 + 5x - 1 = 0 \Rightarrow x = \frac{1}{6}$ ($x = -1$ rejected: LHS negative).

Grade - **High**

Practice map, main concept: Sheet: J-I 16, 19-20; J-II 19-21; J-M 1, 22; J-A 5, 12, 14; C-I 4, 6; C-II 1, 5, 11 | DPP: 6(17); 8(6-10) | Question Bank: 85, 87, 89

Also used in: Sheet: J-I 18; J-II 40, 44; C-I 2 | DPP: 8(5) | Question Bank: 78, 88

Question 43

What to keep in mind while solving ITF equations?

Model Answer

Domain of every term; each side must lie in the range of the ITF on it; taking tan or squaring adds extraneous roots; addition formulas carry conditions ($xy < 1$ etc.); sum of maxima attained only at end points (e.g. $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2} \Rightarrow x = y = z = 1$).

Ex: $\cos^{-1} x + \cos^{-1} 2x = -\pi$ has no solution, since the LHS ≥ 0.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Summation of Series

Question 44

How to solve questions on summation of ITF series?

Model Answer

Write every term as $\tan^{-1}$ ($\cot^{-1} a = \tan^{-1} \frac{1}{a}$, $\sin^{-1}$ via a triangle); find T_r in the form $\tan^{-1} \dfrac{b-a}{1+ab} = \tan^{-1} b - \tan^{-1} a$ (with $b-a$ constant or matching the numerator); sum, so that only the first and last terms remain.

$$\sum_{r=1}^{n} \tan^{-1} \frac{1}{1+r(r+1)} = \tan^{-1}(n+1) - \frac{\pi}{4}, \qquad S_\infty = \frac{\pi}{4}$$

Ex: $\cot\left(\sum_{n=1}^{50} \tan^{-1} \frac{1}{1+n+n^2}\right) = \cot\left(\tan^{-1} 51 - \tan^{-1} 1\right) = \frac{52}{50} = \frac{26}{25}$.

Grade - High

Practice map, main concept: Sheet: J-I 22; J-II 22-23; J-M 5, 16; J-A 1, 6; C-I 3; C-II 6-7 | DPP: 8(11-14) | Question Bank: 73, 83

Also used in: none